

MENTAL HEALTH MONITORING SYSTEM

Phase 3.5 — Severity Head Training on DAIC-WOZ

Progress Report

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1. Overview

Phase 3.5 completes the three-head classifier by training the severity regression head — the component that was frozen throughout Phase 3 due to pending DAIC-WOZ data access. All other components (MentalRoBERTa backbone, GAT layers, emotion head, MH category head) carry forward their trained weights from earlier phases unchanged.

This phase introduces a new training dataset and a new evaluation metric. While Phases 1 and 3 used classification metrics (F1-macro), Phase 3.5 is evaluated using Mean Absolute Error (MAE) on the PHQ-8 scale — a continuous severity score used clinically to measure depression intensity.

Phase objective	Train severity regression head on DAIC-WOZ clinical interviews
New dataset	DAIC-WOZ (AVEC 2017 split) — 107 train / dev participants
Evaluation metric	Mean Absolute Error (MAE) on PHQ-8 scale (0–24)
Published benchmark	MAE $\approx$ 2.85 (best text-only LLM approach on DAIC-WOZ)
Result achieved	MAE = 4.59 Depression F1-macro = 0.7786
Early stopping	Patience = 10 epochs on validation MAE, max 100 epochs
Checkpoints saved	phase35_best.pt, phase35_final.pt, phase35_tokenizer/

2. Why Phase 3.5 Exists

The original Phase 3 plan called for joint training on DepressionEmo and DAIC-WOZ simultaneously. Two blockers prevented this:

- DAIC-WOZ requires a Data Use Agreement (DUA) with USC's Institute for Creative Technologies. Access was pending during Phase 3 execution.
- DepressionEmo was unavailable via HuggingFace at the time of training, so Phase 3 used GoEmotions as a substitute.

Rather than re-running all of Phase 3 once DAIC-WOZ became available, Phase 3.5 takes the Phase 3 checkpoint as its starting point and adds only what was missing: severity head training on DAIC-WOZ transcripts. This is more efficient than restarting from scratch and avoids catastrophic forgetting of the emotion and MH category knowledge already learned.

The severity head was deliberately kept frozen (requires_grad=False) throughout Phase 3 so its random initial weights would not corrupt the gradient signal for the emotion and MH heads during their training. This frozen-then-train strategy is the same approach used in the GAT training phase (Phase 2), where the backbone was frozen while the GAT learned.

3. Dataset — DAIC-WOZ (AVEC 2017)

3.1 Dataset Description

DAIC-WOZ (Distress Analysis Interview Corpus — Wizard of Oz) is the primary clinical benchmark for text-based depression severity estimation. It consists of semi-structured interviews between human participants and a virtual agent named Ellie, controlled by a human operator behind the scenes (the "Wizard of Oz" component).

Dataset	DAIC-WOZ (AVEC 2017 Depression Sub-Challenge)
Source	Kaggle: saifzaman123445/daicwoz
Label format	PHQ-8: continuous score 0–24 + binary (0=not depressed, ≥10=depressed)
Training participants	107
Validation participants	35 (dev split)
Modalities available	Audio, video, facial features, transcripts — text only used here

Transcript format	Tab-separated CSV per participant: timestamp, speaker, utterance
PHQ-8 range (train)	0–23

3.2 Why PHQ-8 and Not PHQ-9

DAIC-WOZ uses PHQ-8 rather than PHQ-9. The ninth PHQ item (suicidal ideation) was excluded from the dataset during original data collection for ethical and safety reasons — participants in a research interview context were not asked directly about suicidal thoughts. This is standard for this benchmark; all published DAIC-WOZ papers report PHQ-8 MAE, making this the correct metric for comparison.

3.3 Data Preparation

Each participant's transcript CSV contains timestamped utterances from both the interviewer (Ellie) and the participant. Only participant speech is retained — interviewer utterances are filtered out before tokenisation. This follows published best practice: Li et al. (2024, IEEE Trans. Computational Social Systems) demonstrated that including interviewer speech causes models to learn shortcuts from question phrasing rather than genuine depression signal from participant responses.

PHQ-8 scores are normalised to the 0–1 range by dividing by 24 before computing MSE loss. Without normalisation, the severity loss operates on a 0–24 scale and produces values up to 576× larger than the cross-entropy MH loss, completely dominating training and preventing the MH head from learning.

Splits follow participant ID boundaries — the same participant never appears in both train and validation. Random splits are explicitly avoided: if split randomly, early and late sessions from the same participant could appear in both sets, leaking conversational context and artificially inflating validation scores.

4. Model Architecture — Phase 3.5

The Phase 3.5 model is structurally identical to Phase 3 with one change: the severity head's parameters have `requires_grad=True`, making it fully trainable.

Layer 1 — Backbone	MentalRoBERTa (768-dim, weights from Phase 1 fine-tuning)
Layer 2 — GAT 1	GATConv: 768 → 4 heads × 256 = 1024-dim (Phase 2 weights)
Layer 2 — GAT 2	GATConv: 1024 → 2 heads → 256-dim (Phase 2 weights)
Head A — Emotion	256 → 128 → 28 outputs, sigmoid (Phase 3 weights, not retrained)
Head B — MH Category	256 → 64 → 2 outputs, softmax (Phase 3 weights, fine-tuned)

Head C — Severity	256 → 128 → 64 → 1 output, Sigmoid (NEW — trained from scratch)
Graph edges	Sequential token edges ($i \rightarrow i+1$), cached per batch shape

4.1 Edge Caching Optimisation

The GAT requires an `edge_index` tensor defining which nodes (tokens) are connected. Rather than recomputing these edges on every forward pass — which would be extremely slow for 256-token sequences across 107 participants — a `_edge_cache` dictionary stores pre-built edge tensors keyed by (`batch_size`, `seq_len`). After the first batch of each size, edge construction cost is zero for all subsequent batches of the same shape.

5. Training Configuration

Severity loss	MSELoss (on normalised 0–1 predictions)
MH category loss	CrossEntropyLoss (binary: depressed / not depressed)
Combined loss	$0.6 \times \text{MSE_severity} + 0.4 \times \text{CE_mh}$
Max epochs	100
Early stopping	Patience = 10 on validation MAE
Batch size	8 (train), 16 (val)
Sequence length	256 tokens
Weight decay	0.01 (L2 regularisation — critical given small dataset)

5.1 Differential Learning Rates

Component	Learning Rate	Reason
Backbone (MentalRoBERTa)	1e-5	Very low — preserve Phase 1+3 knowledge
GAT layers (gat1, gat2)	3e-5	Moderate — gentle adaptation to interview domain
Emotion head	3e-5	Moderate — already trained, light fine-tuning only
MH category head	3e-5	Moderate — adapts to DAIC-WOZ binary labels
Severity head	1e-4	Highest — training from random init, needs to learn fast

5.2 Loss Weighting Rationale

Severity is weighted at 0.6 and MH category at 0.4 because PHQ-8 regression is the primary new objective of this phase. The emotion head is not trained here since DAIC-WOZ has no emotion labels — its loss is masked out by simply not including it in the combined loss computation for DAIC-WOZ batches. This is the masked multi-task loss design described in the project plan.

6. Results

6.1 Key Metrics

Metric	Value	Benchmark / Notes
PHQ-8 MAE (val)	4.59	Published text-only best: ~2.85 (LLM + PHQ indicator extraction)
Depression F1-macro	0.7786	Strong binary classification — consistent with Phase 3 MH head
Training participants	107	Very small clinical dataset — limits regression accuracy
Validation participants	35	AVEC 2017 official dev split
Early stopping epoch	~15–25	Depends on run; MAE plateaued within 25 epochs typically

6.2 Per-Participant Predictions (Sample)

The table below shows the first 10 validation participants, illustrating the model's directional accuracy:

Participant	Predicted PHQ-8	Actual PHQ-8	Error	Clinical Band
1	5.0	4	1.0	None (0–4)→None
2	14.2	4	10.2	None→Moderate
3	11.8	8	3.8	Mild→Moderate
4	9.7	12	2.3	Moderate→Mild
5	8.0	23	15.0	Severe→Mild
6	13.2	19	5.8	Mod.Severe→Moderate
7	9.0	16	7.0	Mod.Severe→Mild
8	10.7	16	5.3	Mod.Severe→Moderate
9	3.2	0	3.2	None→None
10	14.4	17	2.6	Mod.Severe→Moderate

6.3 Result Interpretation

A MAE of 4.59 on the 0–24 PHQ-8 scale means predictions are off by roughly 4–5 points on average — approximately one clinical severity band. The model correctly identifies depression severity direction for most participants but under-predicts high-severity cases (participants 5, 7 with actual scores of 23 and 16 are both under-predicted significantly).

The gap between our MAE of 4.59 and the published best of 2.85 is explained by four factors:

- Dataset size: 107 training participants is extremely small for a regression task. Published LLM results use the same split but leverage models with billions of parameters pretrained on far more clinical text.
- Text only: DAIC-WOZ's strongest published results use audio features (prosody, speech rate, pause patterns) alongside text. Our system is text-only by design.
- General-purpose model: our system is a multi-task model trained on social media text (Reddit), not a DAIC-WOZ-specific model. Domain shift between Reddit posts and clinical interviews contributes to the MAE gap.
- Sequence truncation: interview transcripts are long (often thousands of words) but we truncate to 256 tokens. Some severity signal from later in the interview is lost.

The Depression F1-macro of 0.7786 is strong and suggests the binary classification signal (depressed vs not depressed) transfers well even if the fine-grained severity regression is harder. This is consistent with published literature where binary classification consistently outperforms regression on small clinical datasets.

7. Deliverables

Checkpoint	Location	Purpose
phase35_best.pt	/kaggle/working/	Best val MAE checkpoint — use for inference
phase35_final.pt	/kaggle/working/	Final epoch checkpoint
phase35_tokenizer/	/kaggle/working/	MentalRoBERTa tokenizer for Phase 4

All checkpoints are committed as a Kaggle Output version and persist independently of session timeouts.

8. Overall Project Status

Phase	Status	Notes
Phase 1 — Backbone	✓ COMPLETE	Dreaddit F1=0.7954, GoEmotions F1=0.4876
Phase 2 — GAT Training	✓ COMPLETE	PRIMATE DSM-5 symptom classification
Phase 3 — Three-Head (emotion+MH)	✓ COMPLETE	GoEmotions F1=0.497, MH F1 trained
Phase 3.5 — Severity Head	✓ COMPLETE	MAE=4.59, Depression F1=0.7786
Phase 4 — LLM Integration	PENDING	Groq API + JSON payload → clinical report

9. Next Steps — Phase 4

Phase 4 connects the fully trained classifier to a Large Language Model via the Groq API. No further model training is required. The classifier (phase35_best.pt) produces a structured JSON payload after each inference:

```
{ "emotion": "hopelessness", "emotion_confidence": 0.82,  
  "mh_category": "depression", "severity_score": 0.67,  
  "severity_phq8_equivalent": 16, "risk_flags": [...] }
```

This JSON payload (never the raw user text) is passed to the Groq LLM as structured context in the system prompt. The LLM generates the clinical report, coping strategies, and crisis alerts. The separation between classifier and LLM ensures full auditability — every generated report is traceable back to specific classifier outputs.

- Connect phase35_best.pt inference to JSON serialisation pipeline
- Set up Groq API integration with structured system prompt
- Test end-to-end pipeline: raw text → classifier → JSON → Groq → clinical report
- Validate crisis alert triggering on high-severity / suicidal ideation inputs
- Build demo interface (Gradio or Streamlit) for faculty presentation